A BRIEF SURVEY OF SPACE POLICY AND LAW TOPICS

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1. SPACE LAW AND POLICY

a. Distinction between Private, Public and Commercial International Law

- Commercial International Law deals with transactions between entities in different States;
- Public International Law governs relationship between sovereign states;
- Private International Law is the body of national laws, conventions, model laws, legal guides, and other documents and instruments that determine the governing law applicable to private relationships across national borders.[1] Among other things, it is concerned with *conflicts of laws*, when laws of different jurisdictions might be applicable.

b. Distinction between Policy and Law

- Policy sets overall principles and objectives to be achieved.
- Law provides the framework to implement the policy.

Example: Policy to preserve outer space for all mankind would be implemented by laws prohibiting claims of sovereignty by any State over any part of outer space

Example: Policy to preserve outer space for peaceful uses, implemented by Prevention of Arms Race in Outer Space (PAROS) resolution

c. Primary source of space law is the Outer Space Treaty 1967 (OST), embodying these principles and others reached during the Cold War between US and USSR, each of which aimed to limit the other's appropriation of outer space. Outer space is defined to include celestial bodies.

Article I of OST:

The exploration and use of outer space, including the Moon and other celestial bodies, shall be carried out for the benefit and in the interests of all countries, irrespective of their degree of economic or scientific development, and shall be the province of all mankind.

Outer space, including the Moon and other celestial bodies, shall be free for exploration and use by all States without discrimination of any kind, on a basis of equality and in accordance with international law, and there shall be free access to all areas of celestial bodies.

[1] See American Society of International Law, http://www.asil.org/erg/?page=pil

> There shall be freedom of scientific investigation in outer space, including the Moon and other celestial bodies, and States shall facilitate and encourage international cooperation in such investigation.

Other sources of space law include other treaties on liability and registration, national laws such as UK Outer Space Act 1986, growing body of guidelines and norms.

d. Example of legal dispute involving space: TRW case. TRW proposed the Odyssey satellite system[2] that was not built, but was the subject of a lawsuit and an extraordinary patent claim to the medium earth orbit.[3]

2. ACCESS TO SPACE

Although outer space is vast, the most desirable part is the area where satellites can orbit the Earth to provide services, between the atmosphere and the geostationary orbit (GEO), about 37,000 kilometres above the Earth's surface. The GEO is itself limited and parts of it are in greater demand than others. Therefore, access to these orbits is at a premium and place limits on access by different operators and States. Some factors affecting access are:

a. Orbital slots and frequencies are allocated by the International Telecommunication Union, (ITU). Clearly, some parts of the frequency spectrum are better suited to satellite services than others, introducing further limits to access.

b. The number of launch services available and the window for a particular launch also affect access to space. Therefore, launcher capacity, as well as the availability of insurance can be critical.

3. MICRO AND NANO SATELLITES: ARE THEY DEBRIS?

a. The Inter-Agency Debris Coordination Committee, IADC, Debris Mitigation Guideline defines debris as:

> All man-made objects including fragments and elements thereof, in Earth orbit or re-entering the atmosphere, that are non-functional.[4]

b. Satellites are classified by weight:

APPROXIMATE MAXIMUM WEIGHT

Type	Pounds	Kg
Minisatellite (minisat)	1,100	500kg
Microsatellite (microsat)	220	100kg
Nanosatellite (nanosat)	22	10kg
Picosatellite (picosat)	2.2	1kg

2 See http://ntrs.nasa.gov/archive/nasa/casi.ntrs.nasa.gov/19940018314_1994018314.pdf

3 For background, see http://www.astronautix.com/craft/odyssey.htm and http://personal.ee.surrey.ac.uk/Personal/L.Wood/constellations/overview.html#odyssey For a discussion of some legal ramifications, see http://www.space-institute.org/app/uploads/1362417737_The_TRW_Satellite_Constellation_Claim.pdf

4 IADC Debris Mitigation Guidelines, 2007.

c. With very small satellites, there is less control than larger ones: they don't keep station as well or stay as stable in their orbits. Constant corrections are not possible as fuel is limited. There is no general practice to *passivate* them, as there is with larger satellites. Passivation is the deactivation by complete discharge of onboard energy sources, such as batteries. This is to stop them from exploding if struck by debris. On the other hand, they have less stored energy and are less prone to fragment on being struck.

d. One of the more interesting questions to arise is whether nano- and pico-satellites present the same risks as debris compared to larger satellites. Although they are very small, they are not as well controlled as larger satellites.

4. INTERNATIONAL TRAFFIC IN ARMS REGULATION

ITAR provisions can present an obstacle to active debris removal.

a. ITAR specifies that information and material concerning defence and military related technology (for items on the US Munitions List) may only be shared with US persons, unless authorisation is received from the US Department of State or a special exemption is used. In 2011 the US Department of State issued a rule change to ITAR (section 126.18) that provides an exemption for UK end-user and consignee companies only. However, they have to satisfy some baseline requirements to be approved.

b. Virtually all satellite and space related equipment is on the Munitions List. This covers commercial and non-military dual-use technology.

c. Many satellites have US technology and ITAR prevents "control" of such technology by foreign nationals.

5. RISKS ASSOCIATED WITH SPACE ACTIVITY

a. A brief outline of some areas of risk:

i. Physical and Security, including those associated with the space environment and construction of the space object;

ii. Commercial and Financial, including technology employed;

iii. Spectrum Policy and Regulation, at national, regional and international levels;

iv. Space Policy and Regulation, partly dependent on the nature and manner of implementation of international obligations by national governments; and

v. Insurance, availability and cost.

6. CONCLUSION

Despite the risks, working on space projects and thinking about how to facilitate activities in space is one of the most rewarding and exciting careers anyone could wish for.